Prasant Koirala

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Education

The University of Southern Mississippi – BS in Computer Science, GPA: 3.956 Expected May 2028

- Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Design of Prog. Languages

Experience

Meta, Production Engineering(SRE) Fellow – Remote, San Francisco, CA June 2026

- Selected as one of ∼50 Production Engineering Fellows (SRE) to work on Meta's systems infrastructure — contributing to reliability, observability, and automated deployment of production services.

Cyber Innovation Lab | USM, AI/ML Research Assistant – Hattiesburg, MS Sept. 2025 – Feb. 2026

- Implemented a ModernBERT-based co-attention ML architecture in Python for social engineering detection, achieving a Precision score of 92.2%
- Built data preprocessing pipelines using PyTorch, Pandas, and HuggingFace Transformers from 500,000+ corporate emails, generating balanced sender-receiver pairs across 4 attacker identities

Optimal Answers, Software Engineering Intern – Gulfport, MS Jun. 2025 – Aug. 2025

- Developed REST APIs and C# and SQL backend features for internal and customer-facing web applications, supporting business and operations teams
- Refactored performance-critical backend logic and optimized database queries, reducing UI latency and cutting employee task time by ∼30% across core application flows.
- Designed a staging and pre-release validation environment enabling testing and reducing prod incidents by ∼40%.

Projects

Lavender – Open Source EDR | *Go, Rust, Linux Kernel, Distributed Systems* github.com/Prashant-koi/Lavender

- Built a distributed Linux threat detection system in Go with real-time dashboard, detection workers, and structured event logging for hardware and system diagnostics
- Wrote a Rust eBPF kernel module intercepting Linux syscalls and kernel tracepoints; structured event logs for anomaly detection and hardware diagnostics.
- Implemented NATS-based event streaming as a single source of truth, ensuring ordered, lossless data flow across detection workers and dashboard

No Brainrot – Productivity Browser Extension | *TypeScript, CI/CD* github.com/Prashant-koi/no-brainrot

- Short-form content blocking chrome browser extension across multiple platforms **(289 installs & 20 daily users)**
- Architected modular TypeScript background service worker for the multi-platform content blocker
- Built SPA dashboard for usage analytics and persistent settings using Chart.js and IndexedDB
- Playwright E2E test suite wired into **GitHub Actions** for automated validation on every commit

Curtain – URL Shortener & Analytics Platform | *Python, Flask, PostgreSQL, Docker* github.com/S-Sigdel/Curtain

- Horizontally scaled Flask app tier with Nginx load balancing across two Gunicorn containers; **Redis** split into dedicated roles for caching, short-code allocation, and real-time click tracking
- Production observability with Prometheus, Grafana, structured JSON logs, and Discord webhook alerting; k6 load tests validated **500 concurrent users** at 0.00% error rate and **1.01s p95 latency**.
- Winner of Meta Production Engineering Hackathon with team of 2

Deepseek-go | *Go* github.com/cohesion-org/deepseek-go

- Open source contributor of the Go client of Deepseek with 333 stars

Technical Skills

Languages: Python, Go, Rust, C++, TypeScript
Frameworks: FastAPI, Node.js (Express), React, TailwindCSS
Systems & Infra: Git, Docker, Redis, Nginx, CI/CD (GitHub Actions), GCP, Prometheus, Grafana
Databases: PostgreSQL, MongoDB
AI/ML: PyTorch, Pandas, NumPy, TensorFlow, FAISS, LLM APIs